



THOMAS LABETOUILLE

Gameplay & Tools Programmer — Unreal Engine 5 (C++)

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Driving licence (B) — own vehicle

PROFILE

Software developer combining solid experience in gameplay programming (Unreal Engine 5, C++) with embedded simulator development in the defence sector (C#, Unity). Specialised in editor tooling and pipeline automation: integrating AI agents (Claude AI / MCP) into UE5 workflows, procedural content generation (Editor Scripting, level design), and building anti-regression / CI-CD test suites to make agent-generated code reliable.

PROFESSIONAL EXPERIENCE

Embedded Systems Technician — CS Group

Sep. 2024 – Nov. 2025

- Designed and integrated interactive scenarios for aeronautical simulators (Inscape VTS / national defence), meeting strict functional requirements.
- Developed modular software architectures, debugged complex simulated behaviours, read classified documentation.
- Collaborated with subject-matter experts (systems engineers, pilots) in a regulated defence environment.

Gameplay Programmer (Freelance) — Rainbow Ant Studio

Dec. 2023 – Sep. 2024

- Developed character mechanics, a swimming system, dynamic weather integration and an interactive map in Unreal Engine 5.
- Integrated animations via State Machines and Blend Spaces; wrote progress reports for a remote team.

Personal Project — Agent IA × UE5 — Claude AI / MCP

Nov. 2025 – present

- Built a bridge between an AI assistant (Claude, via MCP) and the Unreal Engine 5 editor: level generation, gameplay mechanic creation and automatic verification of the work produced, all driven from a plain conversation.
- Designed enemy AI: light-based player detection (staying in shadow to avoid being seen), driven by a state machine (patrol, chase, alert, stunned).
- Built a suite of 90 automated tests (CI/CD pipeline, GitHub Actions) that verifies no change breaks the game before it is validated — the same rigor as professional software development.
- Created an autonomous playtest agent able to play the game on its own to test enemy behaviour under real conditions: found and helped fix a real bug preventing enemies from chasing the player.
- Project documented and shared open source on GitHub, with automatic checks on every update (CI) — concrete proof of technical rigor and autonomy (link above).

TECHNICAL SKILLS

Engines & languages:	UE5, Unity, C++, C#, Python, Blueprint (UE5)
AI & Tools:	Claude AI, MCP (Model Context Protocol), UE5 Remote Execution API, AI → UE5 pipeline, Editor Scripting, procedural generation, Behavior Trees, automated testing / CI-CD (GitHub Actions)
Version control:	Git (GitHub, Perforce)
Cross-cutting:	Team management, autonomous remote work, technical English, classified documentation

EDUCATION & CERTIFICATIONS

Bachelor's Degree in Video Game Development — E-Artsup — 2020 – 2023

RNCP Level 6 Certification — Specialisation: C# / Unity / Game & Level Design • Vocational Baccalaureate in Commerce — with distinction

ADDITIONAL INFORMATION

Nationality: French • **Native language:** French

English: fluent technical reading/writing, basic professional spoken (acquired through international work experience)

Interests: Video games, dance, boxing, fitness.